

5. Categorical cross entropy [used in softmax regression]

→ [Multi-classes] {classification}

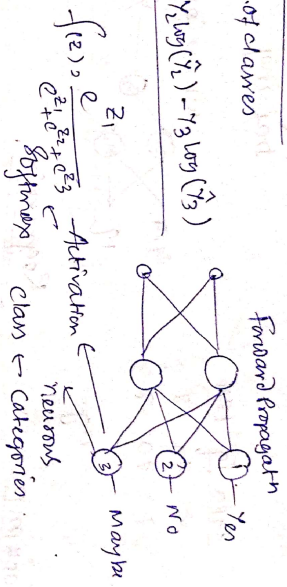
$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

(K = no. of classes)

q _{pr}	i _q	p _{pr}	y _{pr}	N _{pr}	Maya
8	80	70	1	0	0
6	60	60	2	1	0
7	70	70	3	0	1

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j) - y_3 \log(\hat{y}_3)$$

for y_{pr}



for w_{pr}

$$\frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \text{ for Maya} \rightarrow \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Sparse categorical cross entropy

OHE

q _{pr}	i _q	p _{pr}	y _{pr}	N _{pr}	Maya
7	70	70	1	0	1
8	80	80	2	1	0
6	60	60	3	0	1

Integer encode:

$$[0.1, 0.4, 0.5] \rightarrow 1. \text{ Softmax } (y_1 = y_2 = y_3 = 1)$$

$$[0.1, 0.4, 0.5] \rightarrow -y_1 \log(\hat{y}_1)$$

$$[0.1, 0.4, 0.5] \rightarrow -1 \times \log(0.1)$$

$$[0.1, 0.4, 0.5] \rightarrow L \rightarrow (5, 6, 0) \rightarrow -y_3 \log(\hat{y}_3)$$

$$-1 \times \log(0.4)$$

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

$$C = - \frac{1}{n} \sum_{j=1}^n \sum_{i=1}^K y_{ij} \log(\hat{y}_{ij})$$

Summary

Reg → MSE
→ Activations → MAB
→ hidden layers

Classification → binary → binary cross entropy (BCE)
→ multi → multi class cross entropy (MCE)
→ many → (SCE) → sparse

Back propagation

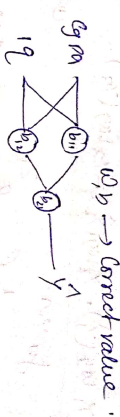
Pre-requisite

→ Gradient descent
→ forward propagation

Backprop
↓
Apply → train NN

q _{pr}	i _q	p _{pr}	y _{pr}	N _{pr}	Maya
8	80	70	1	0	0
6	60	60	2	1	0
7	70	70	3	0	1

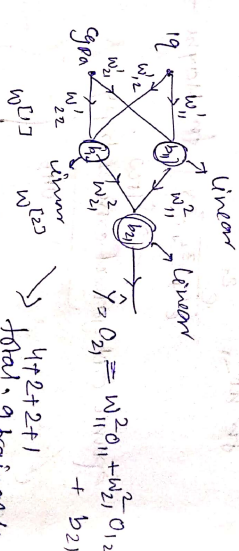
reg.



back prop.

q _{pr}	i _q	p _{pr}	y _{pr}	N _{pr}	Maya
8	80	70	1	0	0
6	60	60	2	1	0
7	70	70	3	0	1

Regression



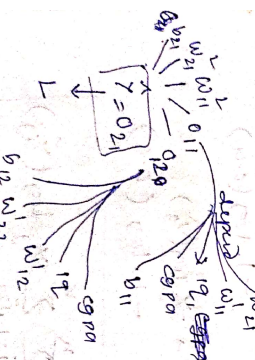
Steps:

1) Init w, b (let $w=1, b=0$)

2) Predict (LPP) → forward propagation

3) choose a loss fn. → MSE

4) w, b - update

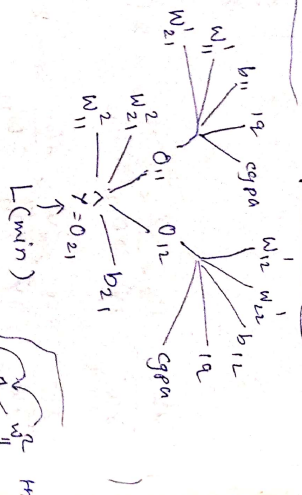
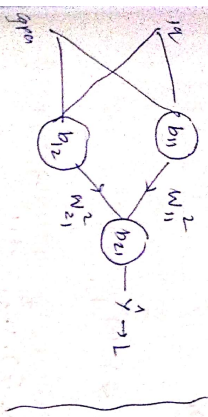


$$w_{old} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

$$w_{old}^2 = w_{old}^2 - \eta \frac{\partial L}{\partial w_{old}^2}$$

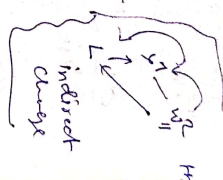
$$b_{old} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}$$

$$\frac{\partial L}{\partial w_{12}^2}, \frac{\partial L}{\partial b_{11}}, \frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{21}^1}, \frac{\partial L}{\partial w_{21}^2}, \frac{\partial L}{\partial b_{12}}, \frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{11}^2}, \frac{\partial L}{\partial b_{11}}, \frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{12}^2}, \frac{\partial L}{\partial b_{12}}$$



You cannot find $\frac{\partial L}{\partial w_{12}^2}$ directly. !!

You have to find out $\rightarrow \frac{\partial L}{\partial w_{12}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{12}^2}$



$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (y - \hat{y})^2 = -2(y - \hat{y})$$

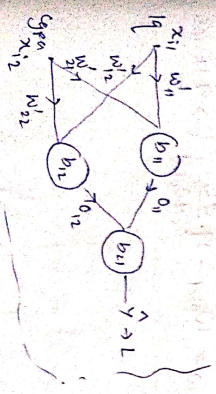
$$\frac{\partial \hat{y}}{\partial w_{12}^2} = \frac{\partial}{\partial w_{12}^2} [0_{11} w_{11}^2 + 0_{12} w_{21}^2 + b_{21}] = 0_{11}$$

$$\therefore \frac{\partial L}{\partial w_{12}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{12}^2} = -2(y - \hat{y}) \cdot 0_{11}$$

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{21}^2} = -2(y - \hat{y}) \cdot 0_{12}$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{21}} = -2(y - \hat{y}) \cdot 1 = -2(y - \hat{y})$$

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{11}} \times \frac{\partial 0_{11}}{\partial w_{11}^1} = -2(y - \hat{y}) \cdot w_{12}^2 \cdot x_{11}$$



$$\frac{\partial L}{\partial w_{21}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial w_{21}^1} = -2(y - \hat{y}) \cdot w_{11}^2 \cdot 1$$

$$\frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{11}} \times \frac{\partial 0_{11}}{\partial b_{11}} = -2(y - \hat{y}) \cdot w_{12}^2 \cdot 1$$

$$\frac{\partial L}{\partial w_{11}^1} = -2(y - \hat{y}) \cdot w_{12}^2 \cdot x_{11}$$

$$\frac{\partial L}{\partial w_{12}^1} = -2(y - \hat{y}) \cdot w_{11}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y}) \cdot w_{12}^2$$

$$\frac{\partial L}{\partial w_{12}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial w_{12}^2} = -2(y - \hat{y}) \cdot w_{21}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{11}} \times \frac{\partial 0_{11}}{\partial w_{21}^2} = -2(y - \hat{y}) \cdot w_{11}^2 \cdot x_{11}$$

$$\frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial 0_{12}} \times \frac{\partial 0_{12}}{\partial b_{12}} = -2(y - \hat{y}) \cdot w_{21}^2$$

$$\frac{\partial L}{\partial w_{12}^1} = -2(y - \hat{y}) \cdot w_{21}^2 \cdot x_{12}$$

$$\frac{\partial L}{\partial w_{21}^1} = -2(y - \hat{y}) \cdot w_{11}^2 \cdot x_{11}$$

$$\frac{\partial L}{\partial b_{12}} = -2(y - \hat{y}) \cdot w_{21}^2$$

Step 1: Weights, bias $\rightarrow$ random

Step 2: for i in range (4):

1) ai + student $\rightarrow$ forward propagation

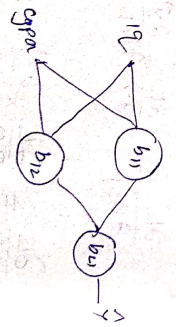
predict (LPR)

2) loss calculate (MSE)

3) adjust all weights and bias

(new w, b)

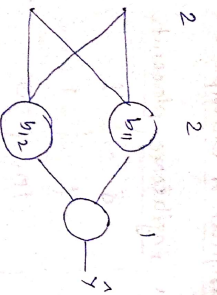
$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{\partial L}{\partial w_{\text{old}}} \rightarrow \text{9 times}$$



epoch	1	2	3	4
loss	8.50	6.00	7.00	7.00

Regression problem

epoch	train score	test score
8	8	4
7	9	5
6	10	6
5	12	7



Backpropagation Algorithm

epochs = 5

for i in range(epochs):

for j in range(X.shape[0]):

→ select 1 row (random)

→ predict (using forward propagation)

→ calculate loss (using loss fn → MSE)

→ update weights & bias using GD

$$w_1 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

→ calculate avg loss for the epoch

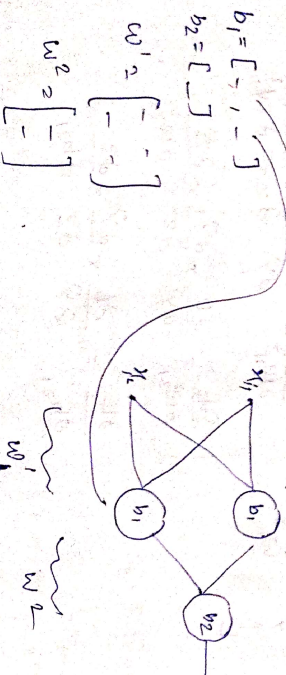
$$\frac{\partial L}{\partial w_{11}} = -2(y - \hat{y})w_{11}^2x_{11}$$

$$\frac{\partial L}{\partial w_{12}} = -2(y - \hat{y})w_{11}^2x_{12}$$

$$\frac{\partial L}{\partial w_{21}} = -2(y - \hat{y})w_{11}^2x_{12}$$

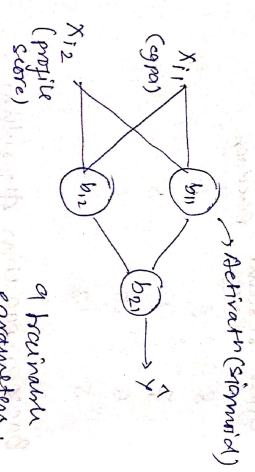
$$\frac{\partial L}{\partial w_{22}} = -2(y - \hat{y})w_{11}^2x_{12}$$

Regression problem



Classification example

epoch	train score	test score
8	8	1
7	9	1
6	10	0
5	12	0



Backpropagation algorithm

epochs = 5

for i in range(epochs):

for j in range(X.shape[0]):

→ select 1 row (random)

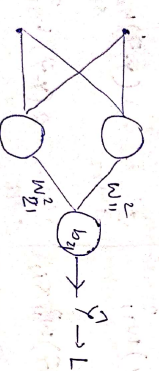
→ predict (using forward prop)

→ calculate loss (using loss function → BCE)

→ update w, b using GD

$$w_1 = w_0 - \eta \cdot \frac{\partial L}{\partial w_0} \rightarrow \text{grad}$$

→ calculate avg loss for the epoch



$$L = -\gamma \log(\hat{y}) - (1-\gamma) \log(1-\hat{y})$$

$$\frac{\partial L}{\partial w_{11}} = w_{11}^2x_{11} + w_{21}^2x_{12} + b_{21}$$

$$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial w_{12}}$$

$$\frac{\partial L}{\partial w_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial w_{21}}$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial b_{21}}$$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} \left[-y \log(\hat{y}) - (1-y) \log(1-\hat{y}) \right]$$

$$= -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})} (-1) = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{(1-\hat{y})^2}$$

$$= \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{(1-\hat{y})^2} = \frac{\hat{y} - y}{(1-\hat{y})^2}$$

$$\hat{y} = \sigma(z)$$

$$= \frac{-\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})}}{(1-\hat{y})^2}$$

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial}{\partial z} \sigma(z) = \sigma(z) [1 - \sigma(z)]$$

$$= \hat{y} (1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})} \cdot \frac{\partial \hat{y}}{\partial z} = \hat{y} (1 - \hat{y})$$

$$\left\{ \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} = -(y - \hat{y}) \right\}$$

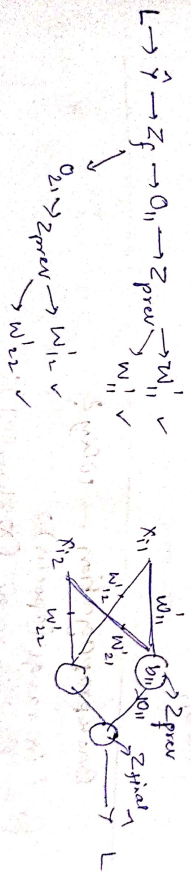
$$\frac{\partial z}{\partial w_{11}} = 0_{11}$$

$$\frac{\partial L}{\partial w_{11}^2} = -(y - \hat{y}) \cdot 0_{11}$$

$$\frac{\partial L}{\partial w_{21}} = -(y - \hat{y}) \cdot 0_{12}$$

$$\frac{\partial L}{\partial b_{21}} = -(y - \hat{y})$$

$$\{ z = w_{11}^2 0_{11} + w_{21}^2 0_{12} + b_{21} \}$$



$$\frac{\partial L}{\partial w_{11}^2} = \left[\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \right] \times \left[\frac{\partial z}{\partial w_{11}^2} \right] \times \left[\frac{\partial z}{\partial z_{prev}} \right] \times \left[\frac{\partial z_{prev}}{\partial w_{11}^2} \right]$$

$$= (y - \hat{y}) \cdot w_{11}^2 \cdot 0_{11} \cdot (1 - 0_{11})$$

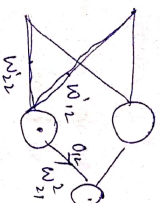
$$z_{prev} = x_{11} w_{11}^1 + x_{12} w_{21}^1 + b_{11}$$

$$\frac{\partial z_{prev}}{\partial w_{11}^1} = x_{11}$$

$$\frac{\partial L}{\partial w_{11}^1} = -(y - \hat{y}) \cdot w_{11}^2 \cdot 0_{11} \cdot (1 - 0_{11}) \cdot x_{11}$$

$$\frac{\partial L}{\partial w_{21}^1} = -(y - \hat{y}) \cdot w_{11}^2 \cdot 0_{11} \cdot (1 - 0_{11}) \cdot x_{12}$$

$$\frac{\partial L}{\partial b_{11}} = -(y - \hat{y}) \cdot w_{11}^2 \cdot 0_{11} \cdot (1 - 0_{11})$$



$$\frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{22}^1}, \frac{\partial L}{\partial b_{12}}$$

$$z_1 \rightarrow z_2 \rightarrow 0_{12}$$

$$\frac{\partial L}{\partial w_{12}^1} = \left[\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \right] \times \left[\frac{\partial z}{\partial w_{12}^1} \right] \times \left[\frac{\partial z}{\partial z_{prev}} \right] \times \left[\frac{\partial z_{prev}}{\partial w_{12}^1} \right]$$

$$\frac{\partial L}{\partial w_{12}^1} = -(y - \hat{y}) \cdot w_{21}^2 \cdot 0_{12} \cdot (1 - 0_{12}) \cdot x_{11}$$

$$\frac{\partial L}{\partial w_{22}^1} = -(y - \hat{y}) \cdot w_{21}^2 \cdot 0_{12} \cdot (1 - 0_{12}) \cdot x_{12}$$

$$\frac{\partial L}{\partial b_{12}} = -(y - \hat{y}) \cdot w_{21}^2 \cdot 0_{12} \cdot (1 - 0_{12})$$

Backpropagation - why?

for i in range(epochs):
for j in range(X.shape[0]):

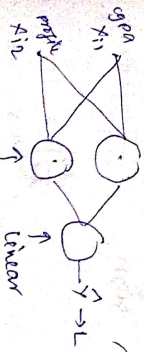
- select f and (random)
- predict (using forward propagation)
- calculate loss (using loss-function → mse)
- update weights and biases - GD

$$w_n = w_0 - \eta \left[\frac{\partial L}{\partial w_0} \right] \rightarrow \text{magic} !!$$

calculate avg loss for the epoch

→ Loss fn is a function of all trainable parameters

eg for 2 inputs 1 output



$$MSE = (y - \hat{y})^2$$

$$L = (y - \hat{y})^2$$

$$L(?)$$

Mathematical function

linear

$$\hat{y} = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_1$$

$$\hat{y} = w_{11}^2 [w_{11}^1 x_{11} + w_{12}^1 x_{12} + b_{11}] + w_{21}^2 [w_{11}^1 x_{11} + w_{12}^1 x_{12} + b_{12}] + b_2$$

$$L(\hat{y}) = L(w, b)$$

all the weights & biases

→ Concept of Gradient

Gradient descent

$$y = f(x) = x^2 + x$$

$$\frac{dy}{dx} = \frac{d}{dx} (x^2 + x) = \frac{d}{dx} (x^2 + x) = 2x + 1$$

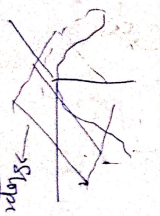
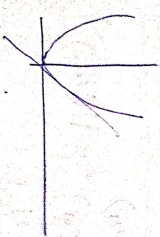
derivative $y \rightarrow x \rightarrow$ only one parameter $\rightarrow$ derivative $\cdot \frac{d}{dx}$

$$L(w, b) \rightarrow L(w_{11}^1, w_{12}^1, \dots, w_{21}^2, b_{11}, b_{12}, b_2)$$

9 parameters

$$y = f(x) = x^2 + x$$

$$L(w, b) \rightarrow \left[\frac{\partial L}{\partial w}, \frac{\partial L}{\partial b} \right]$$



$$z = x^2 + y^2$$

$$z = f(x, y)$$

Calculating 9 different slopes w.r.t. each dim.

→ Concept of derivative

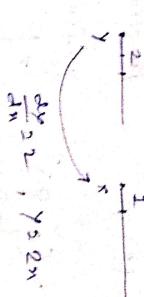
derivative at a point

$\frac{dy}{dx}$ rate of change

$$y = \text{func } x \rightarrow \frac{dy}{dx} = 2(+ve) \text{ sign}$$

$$y = x^2 + 2x \rightarrow \frac{dy}{dx} = (2x + 1)_{x=5}$$

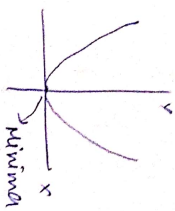
$$\frac{dy}{dx} \Big|_{x=5} = 11 \text{ slope}$$



$$\left[\frac{\partial L}{\partial w_{11}^1} \right]$$

$w_{11}^1 \rightarrow$ linear curve $L \rightarrow ?$ (sign)

→ The concept of Minimum



$$\left(\frac{dy}{dx} \right) = 0$$

$x = ?$ minima

$$z = x^2 + y^2$$

$$\frac{\partial z}{\partial x} = 2x = 0 \rightarrow x = 0$$

$$\frac{\partial z}{\partial y} = 2y = 0 \rightarrow y = 0$$

minima $\rightarrow (0, 0)$



→ Backpropagation intuition

$$w_{11} = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$\eta = 1$ 9 steps

$$w_{11} = w_0 - \eta \frac{\partial L}{\partial w_0}$$

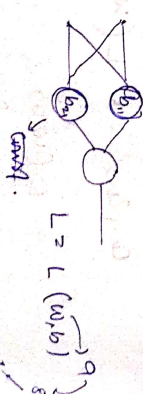
$$b_{21} = b_{21} - \eta \frac{\partial L}{\partial b_{21}}$$

$$\frac{\partial L}{\partial b_{21}}$$

L derivative w.r.t. b_{21}

$$\frac{\partial L}{\partial b_{21}} = +ve \rightarrow b_{21} \uparrow$$

$\rightarrow$ so we have to reduce b_{21} $\rightarrow$ so we have to $\uparrow b_{21}$



Cont.

$$L = (y - \hat{y})^2 \rightarrow y \rightarrow (b_{21}) \rightarrow L \rightarrow (b_{21}) = L(b_{21})$$

